



AI SMART AGRIBOT

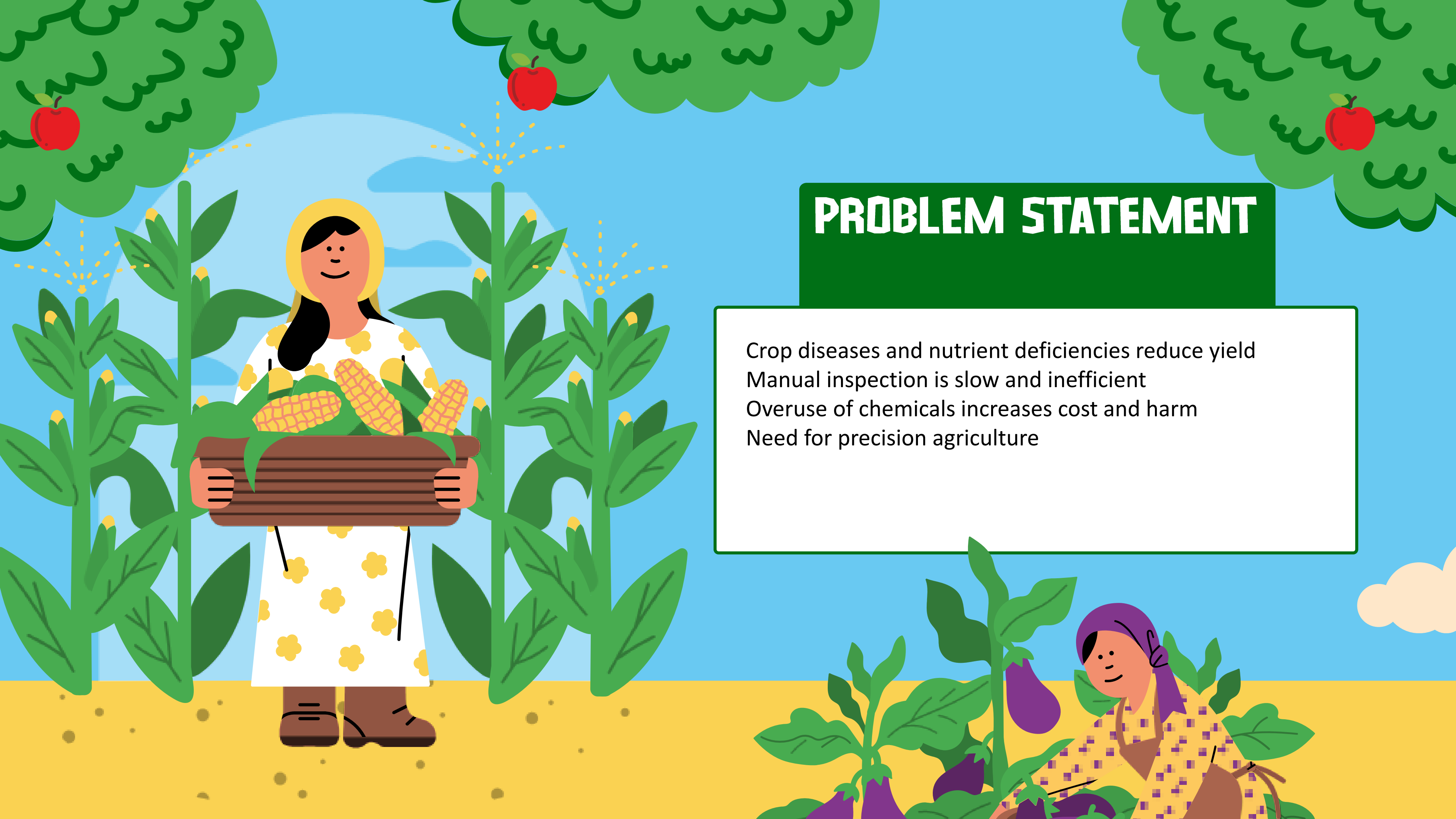
WITH SMART ROTATING SPRAY SYSTEM

CHAMOD MINIDU KAHNDSE25ZF-005

SHADEESHA LAKMAL KAHNDSE25ZF-014

MADUSHA RAJAPAKSHA KAHNDSE25ZF-031

DINETH WEERASEKARA KAHNDSE25ZF-032



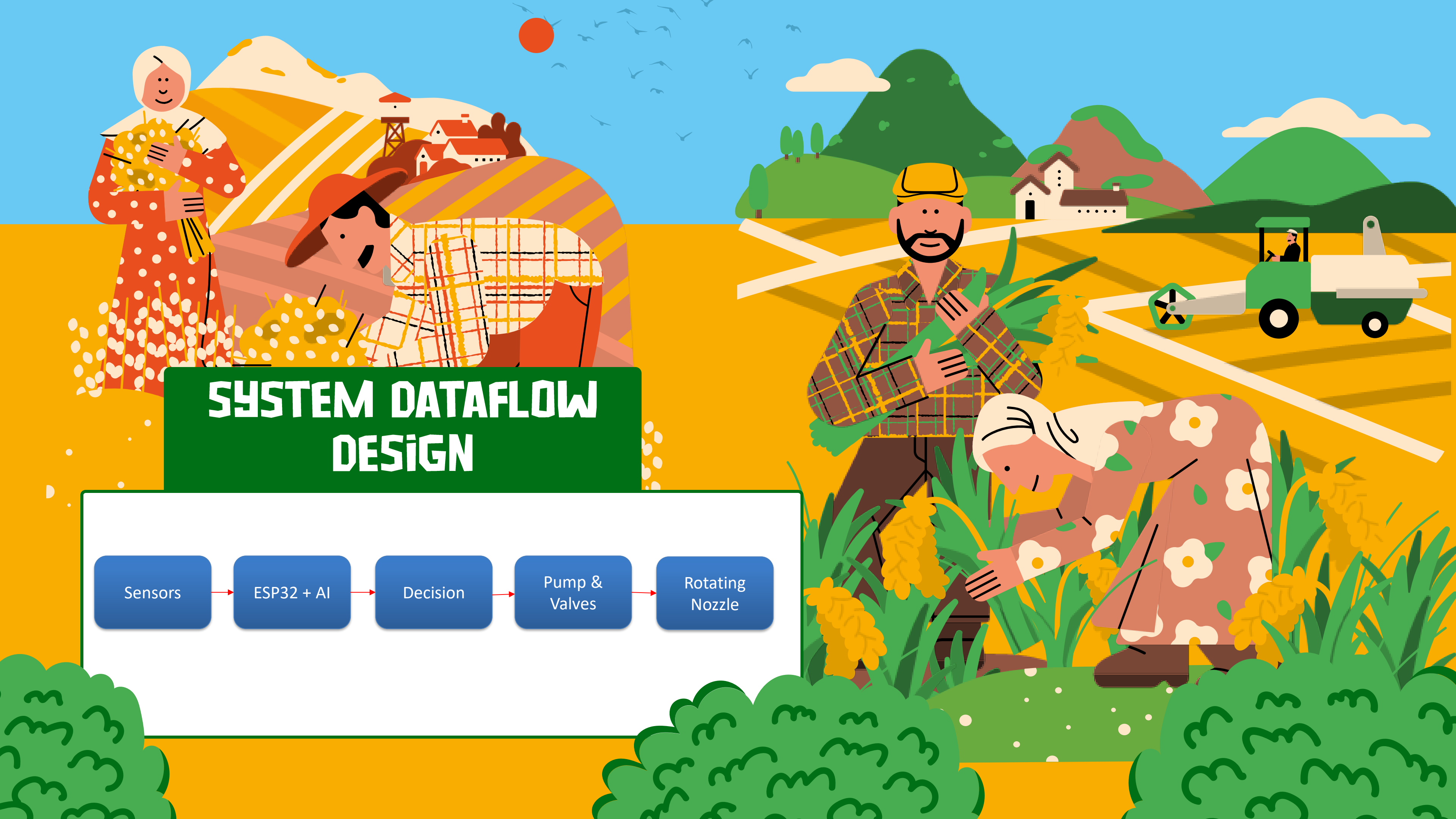
PROBLEM STATEMENT

Crop diseases and nutrient deficiencies reduce yield
Manual inspection is slow and inefficient
Overuse of chemicals increases cost and harm
Need for precision agriculture

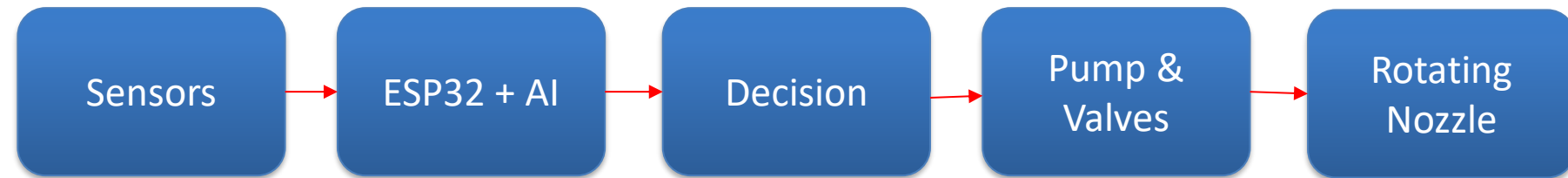
PROPOSED SOLUTIONS

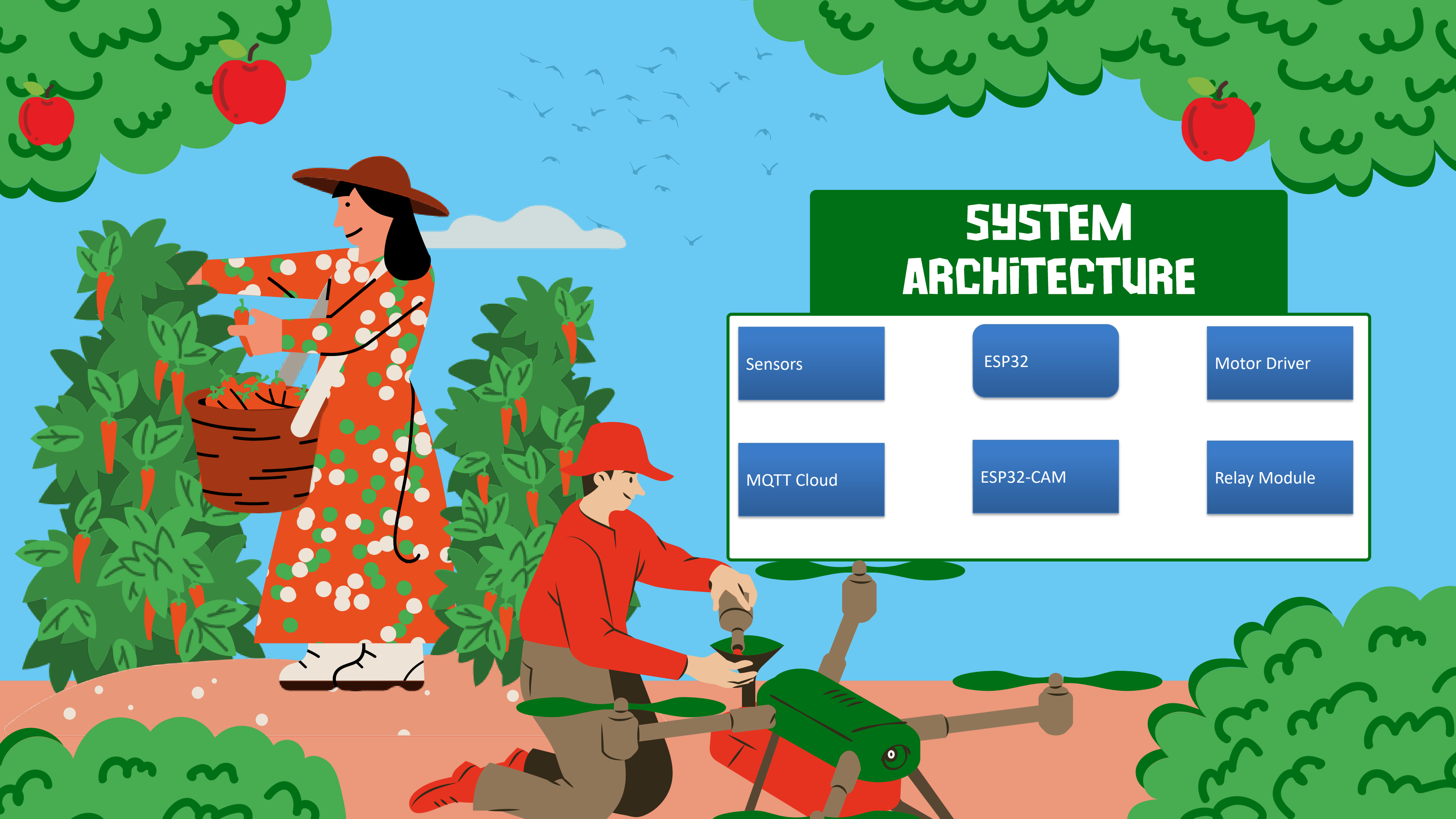
Autonomous robot for field monitoring
AI-based leaf analysis using ESP32-CAM
Selective spraying via rotating nozzle head
Real-time alerts via MQTT





SYSTEM DATAFLOW DESIGN





SYSTEM ARCHITECTURE

Sensors

ESP32

Motor Driver

MQTT Cloud

ESP32-CAM

Relay Module

A colorful illustration of a farm scene. In the foreground, a man in a brown hat and orange vest holds a large watermelon. To his left, another man in a plaid shirt is bent over, working in a field. In the bottom left, a woman in a yellow headscarf and floral dress is harvesting lemons from a tree. The background features rolling green hills, a red tractor pulling a trailer, a white water tower, and a red barn. The sky is blue with white clouds and small birds. A large green bush is in the bottom center, and a red apple hangs from a tree in the top left.

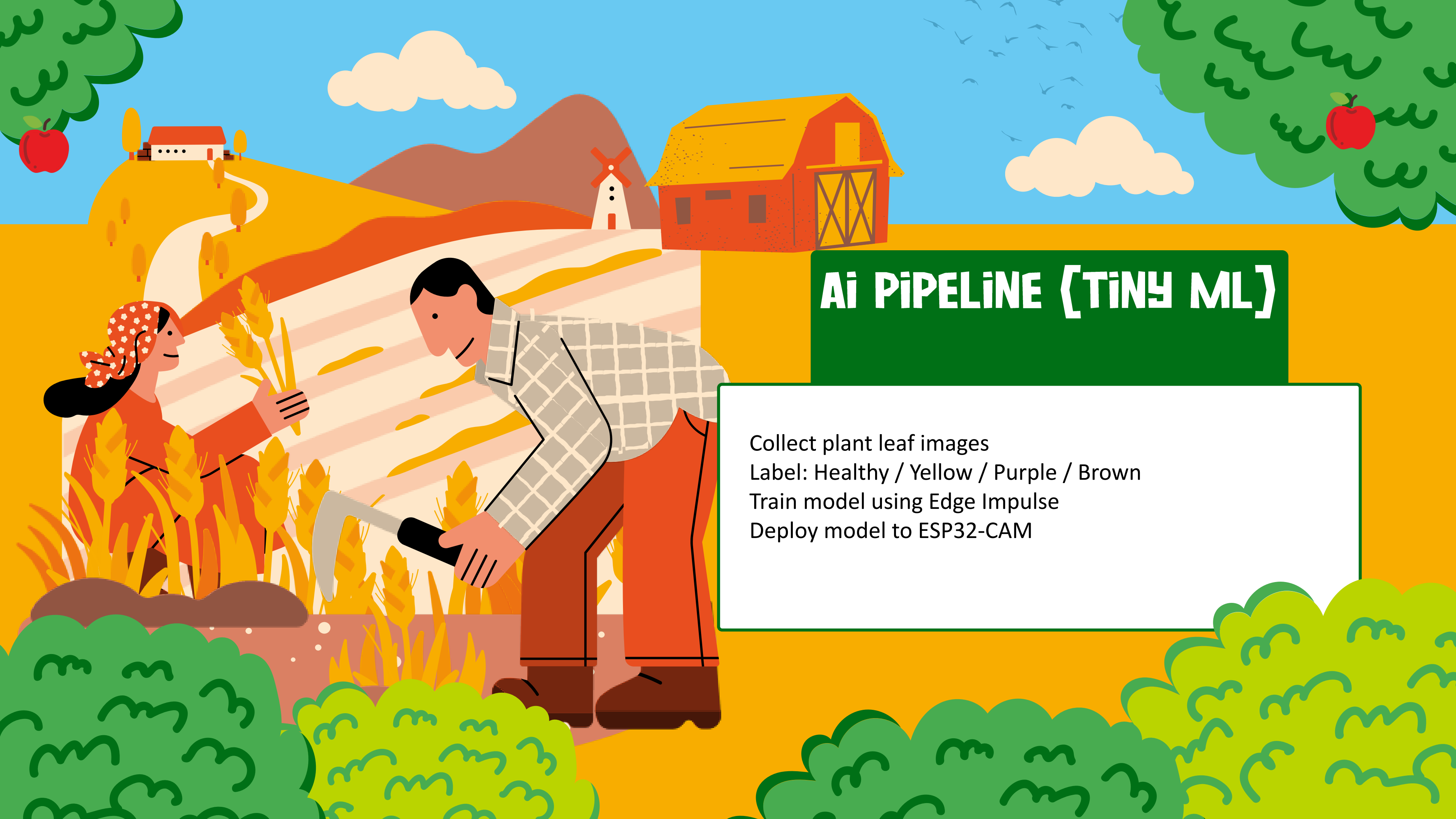
HARDWARE DESIGN

4WD robot chassis with DC motors

4 chemical tanks + 1 water tank

Pump + solenoid valves

Dual-axis servo (rotation + tilt) nozzle system



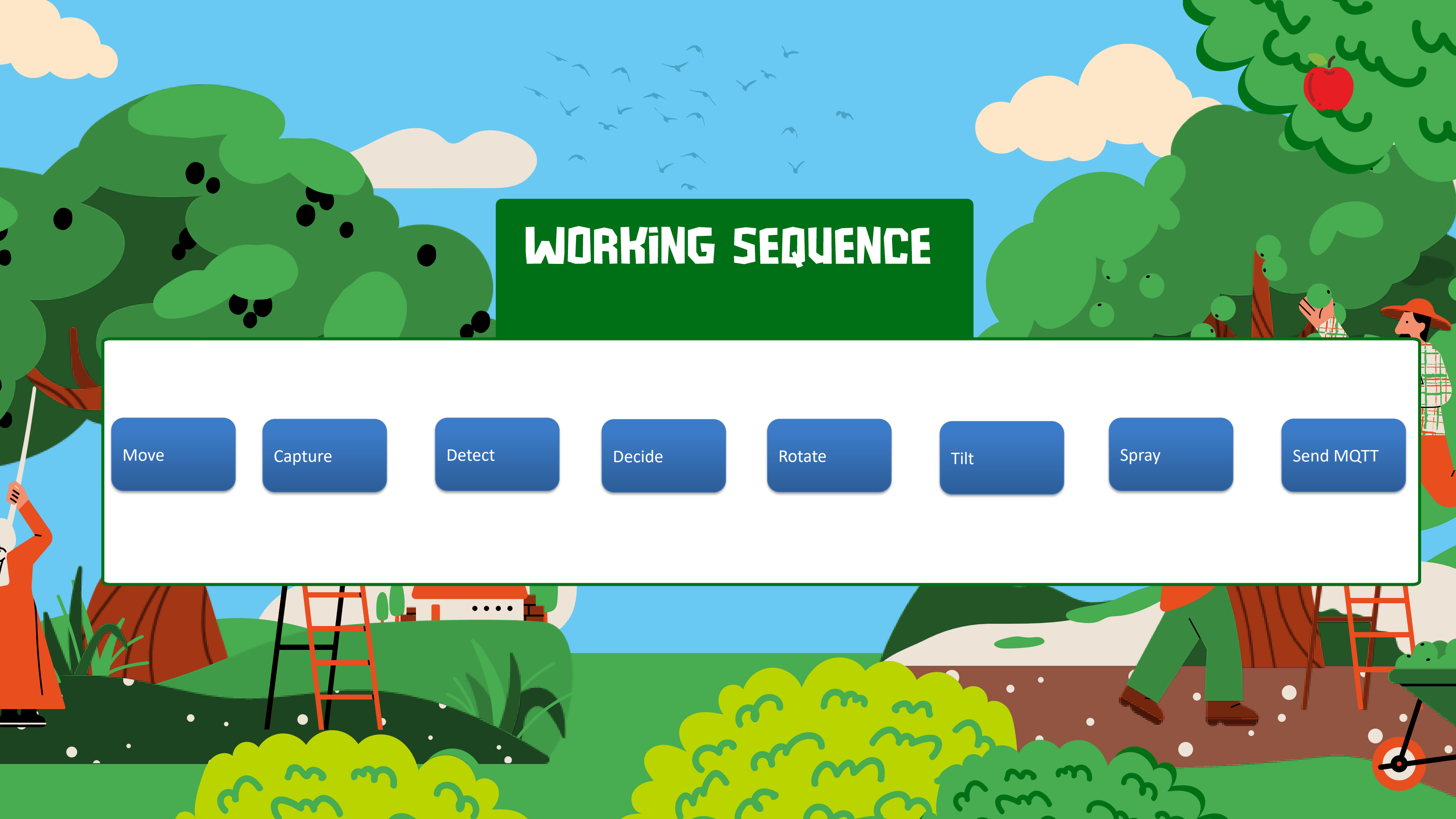
AI PIPELINE (TINY ML)

Collect plant leaf images

Label: Healthy / Yellow / Purple / Brown

Train model using Edge Impulse

Deploy model to ESP32-CAM

The background is a vibrant, stylized illustration of a farm. On the left, a person in an orange coat stands near a tree with black berries. In the center, a green sign with white text reads 'WORKING SEQUENCE'. To the right, another person in a red hat and plaid shirt is near a tree with green fruit. The sky is blue with many small birds flying. The ground is green with yellow bushes in the foreground. A ladder and a wheelbarrow are also visible.

WORKING SEQUENCE

Move

Capture

Detect

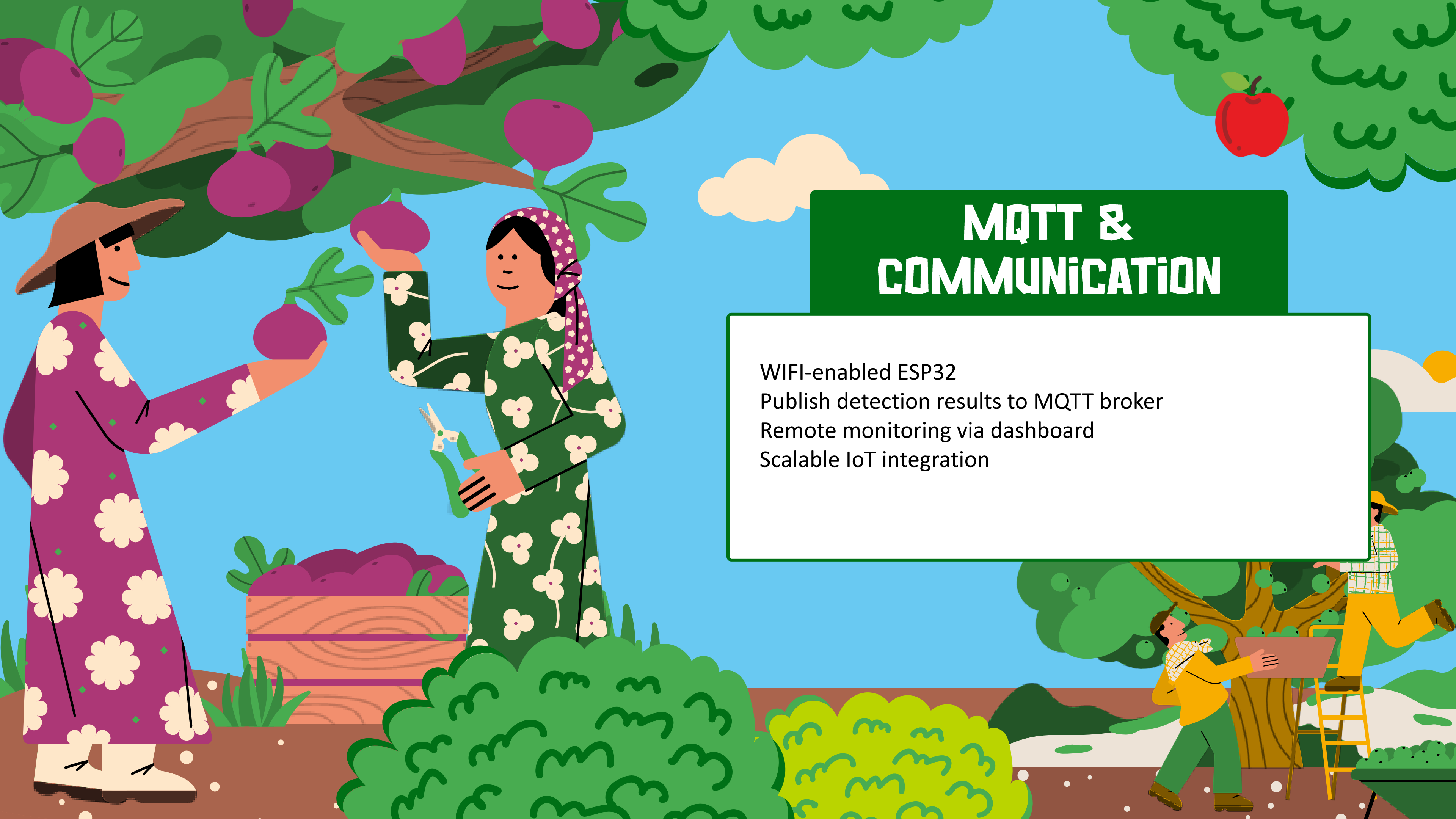
Decide

Rotate

Tilt

Spray

Send MQTT



MQTT & COMMUNICATION

WIFI-enabled ESP32
Publish detection results to MQTT broker
Remote monitoring via dashboard
Scalable IoT integration

BUDGET & EXPECTED OUTCOMES

Total Cost: LKR 25,000
Reduced chemical wastage
Improved crop health monitoring
Scalable smart farming solution



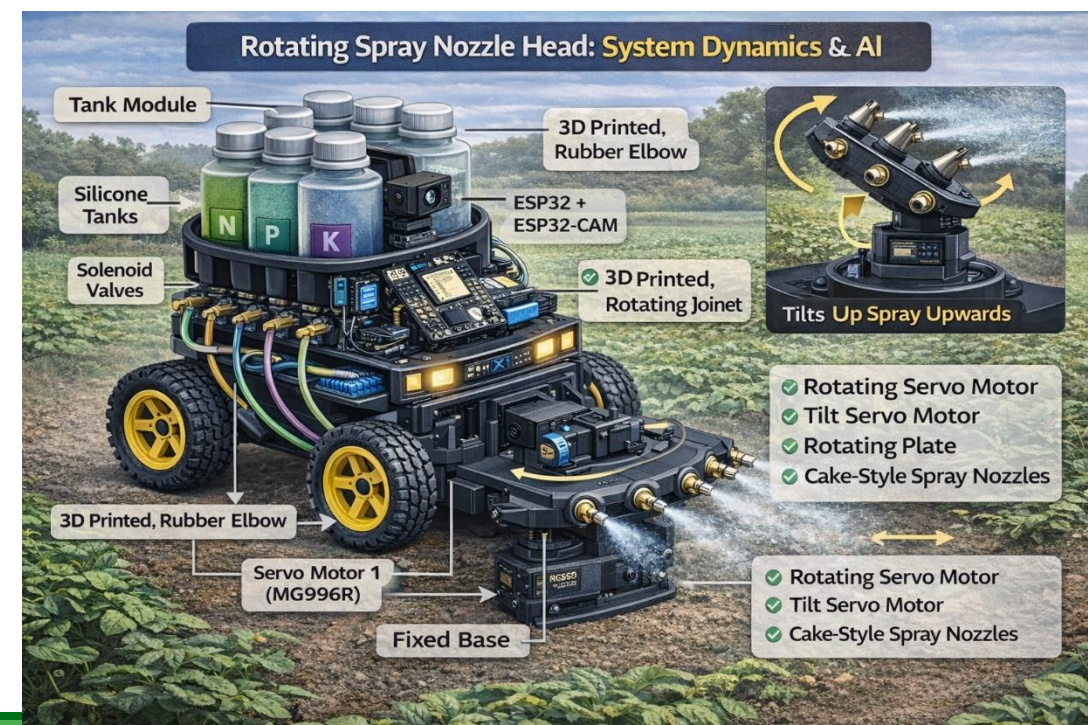
SYSTEM

The robot has four wheels and carries four chemical tanks on top labeled nitrogen, phosphorus, potassium, and water.

At the front, there is a rotating spray nozzle system with four metal cake-style nozzles mounted on a circular plate.

The nozzle head rotates clockwise smoothly while also tilting upward using a dual-axis servo mechanism. Flexible tubes connect the tanks to the nozzles.

The robot scans plant leaves using a small camera (ESP32-CAM). When a yellow leaf is detected, the robot stops, the nozzle head rotates toward the plant, tilts upward, and sprays liquid precisely.





Meet the future of precision farming: an



THANK
YOU

